CivicFix AI — AI-Powered Urban Infrastructure Complaint Prioritization

Assignment 1 · Project Proposal · Group 8: Tunahan Onur Direk, Yağmur Sert, Sezin Naz Sarı
GitHub repository: https://github.com/yagmursertt/civicfix-ai

1. SDG and target

SDG 11 — Sustainable Cities and Communities, Target 11.5: significantly reduce the number of deaths and people affected, and the economic losses, caused by disasters, including water-related disasters. **Supporting:** SDG 9, Target 9.1 (quality, reliable and resilient infrastructure) and Target 11.7 (safe and accessible public spaces, notably for older people and persons with disabilities).

2. Problem, target users and the tool in one sentence

Problem. Municipalities receive thousands of citizen complaints about potholes, cave-ins, broken sidewalks, missing manhole covers and clogged storm drains, mostly as free text. They are handled largely by category and arrival order, so dangerous defects can wait behind minor ones, and blocked drains are often noticed only when streets flood. On 14–15 October 2026 Istanbul received up to 273.1 kg/m² of rain (Beykoz); İBB logged 8,515 emergency calls, 16 roads were closed, walls and a bridge wall collapsed, and at least six people died [2, 3].

Target users. Municipal maintenance and operations planners (road maintenance, sewer and drainage units, district municipalities) who decide daily which reported problems crews visit first. Citizens benefit indirectly through faster repair of the most dangerous defects.

In one sentence: *CivicFix AI helps municipal maintenance planners decide which reported road, sidewalk and drainage problems to fix first, by turning free-text citizen complaints into a risk-ranked work queue that accounts for hazard, recurring hotspots and rainfall.*

How we will know it works. (i) the risk model ranks future problem spots better than arrival order (ROC-AUC, precision in the top 10%); (ii) the LLM classifies complaints correctly (macro-F1 on a hand-labelled set); (iii) more Critical issues reach the top of the queue than with first-come, first-served ordering.

3. Supporting data source

NYC 311 Service Requests (NYC Open Data, erm2-nwe9) [1] — the largest open, record-level municipal complaint log. We use all 2025 Street, Sidewalk and Sewer complaints: 132,653 records with issue type, time, location and closing date (daily rainfall from NOAA [4] is joined by date). The data supports the problem: 18.8% of complaints are Critical hazards; on days with >20 mm of rain, flood-related complaints rise from 42 to 237 per day (≈5.7×); cave-ins wait a median 2.3 days and collapsed sidewalks 3.4 days, longer than potholes (0.9 days); and the 10% most-reported ~110 m cells hold 33.7% of all complaints. Comparable Turkish data is only published as district totals, so we train on NYC and adapt to Turkish complaints through the LLM step.

4. Flow diagram — what goes in, what comes out

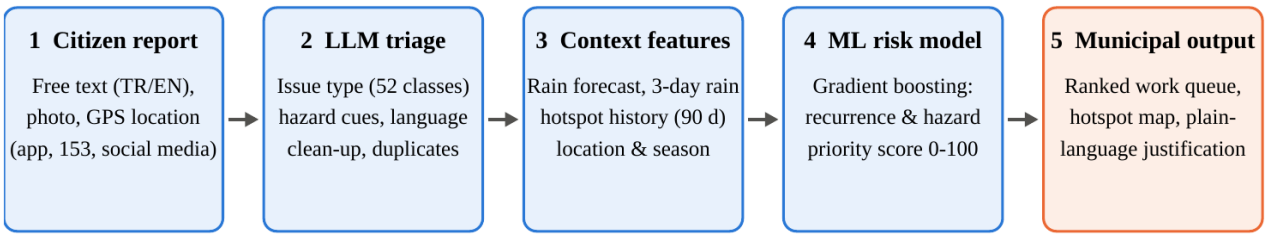


Figure 1. The user (citizen / call-centre operator) enters a free-text complaint with location; the planner receives a ranked queue, a hotspot map and a one-line justification. Steps 2 and 4 are the AI models.

5. Models and why

(a) LLM for complaint triage. Turkish complaints are short and informal, and no labelled Turkish training data exists. A general-purpose LLM, called through an API with a structured JSON prompt, can classify each text zero-shot into our 52 issue types and three hazard tiers (Critical / High / Routine) and give a short reason. Low-confidence outputs go to human review; the LLM never closes a complaint.

(b) Gradient boosting for risk. Predicts whether the same location will be reported again within 30 days (a recurring hotspot) from issue type, hazard tier, location type, season, rainfall and past complaints at that spot. Chosen because it handles mixed categorical/numeric features, captures non-linear rain effects, trains fast on hundreds of thousands of rows and is explainable (feature importance/SHAP), which matters for a public-sector tool. **Priority score** = hazard weight × (1 + recurrence probability) × rain factor for drainage issues.

6. First experiments and results

After cleaning (duplicates, unknown boroughs, invalid closing times; 132,511 records), we mapped issues to hazard tiers, snapped locations to a ~110 m grid, and built features using only information available when the complaint arrives. Four models were compared with a chronological split:

Model	ROC-AUC	PR-AUC	F1
Baseline (majority)	0.500	0.399	0.000
Logistic Regression	0.670	0.586	0.540
Random Forest	0.673	0.584	0.571
Hist. Gradient Boosting	0.672	0.584	0.561

Table 1. Recurring-hotspot prediction; train Jan–Aug 2025, test Sep–Nov 2025 (27,940 complaints, 39.9% positive).

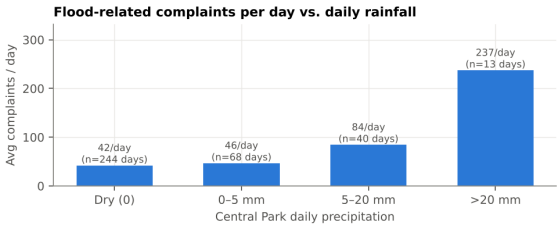


Figure 2. Flood complaints per day vs. rainfall (EDA).

All learned models clearly beat the baseline (ROC-AUC ≈ 0.67 vs 0.50; PR-AUC ≈ 0.58 vs 0.40). If crews inspect the 10% highest-scored complaints first, **71.6%** are confirmed recurring hotspots, against a 39.9% base rate. Gradient boosting is kept as the main model; next we will add neighbour-cell and rain-forecast features, tune it, and build and validate the LLM triage on ~200 hand-labelled Turkish complaints (Assignment 2). **Ethics note:** areas that report less must not be ranked lower — we will audit rankings by district and strip personal data before any API call.

References

[1] NYC Open Data, "311 Service Requests from 2010 to Present" (erm2-nwe9), accessed 19 Oct 2026. [2] Hürriyet, "İstanbul'u sel vurdu…", live coverage, 15 Oct 2026 (İBB AKOM rainfall and call figures). [3] Diken, "İstanbul'da sel: Can kaybı altıya yükseldi", 19 Oct 2026. [4] NOAA NCEI, GHCN-Daily, station USW00094728 (Central Park, NY). [5] United Nations, SDG targets 9.1, 11.5, 11.7 (sdgs.un.org).